Dead space

Some of the inspired air never reaches the gas exchange areas but fills the respiratory passages where exchange of gases does not occur. This air space is called dead space.

Dead space is not involved in gaseous exchange. It amounts to approximately 150mL.

6.4 Exchange of gases

The primary site for the exchange of gases is the alveoli. The uptake of O_2 and the release of CO_2 occur between the blood and tissues by simple diffusion driven by partial pressure gradient of O_2 and CO_2. Partial pressure is the pressure contributed by an individual gas in a mixture of gases. It is represented as pO_2 for oxygen and pCO_2 for carbon–dioxide. Due to pressure gradients, O_2 from the alveoli enters into the blood and reaches the tissues. CO_2 enters into the blood from the tissues and reaches alveoli for elimination. As the solubility of CO_2 is 20–25 times higher than that of O_2, the partial pressure of CO_2 is much higher than that of O_2 (Table 6.1 and Figure 6.6).

Respiratory pigments

Haemoglobin

Haemoglobin belongs to the class of conjugated protein. The iron containing pigment portion haem constitutes only 4% and the rest colourless protein globin belongs to histone class. Haemoglobin has a molecular weight of 68,000 daltons and contains four atoms of iron, each of which can combine with a molecule of oxygen.

Methaemoglobin

If the iron component of the haem is in the ferric state, than the normal ferrous state, it is called methaemoglobin. Methaemoglobin does not bind with O_2. Normally RBC contains less than 1% methaemoglobin.

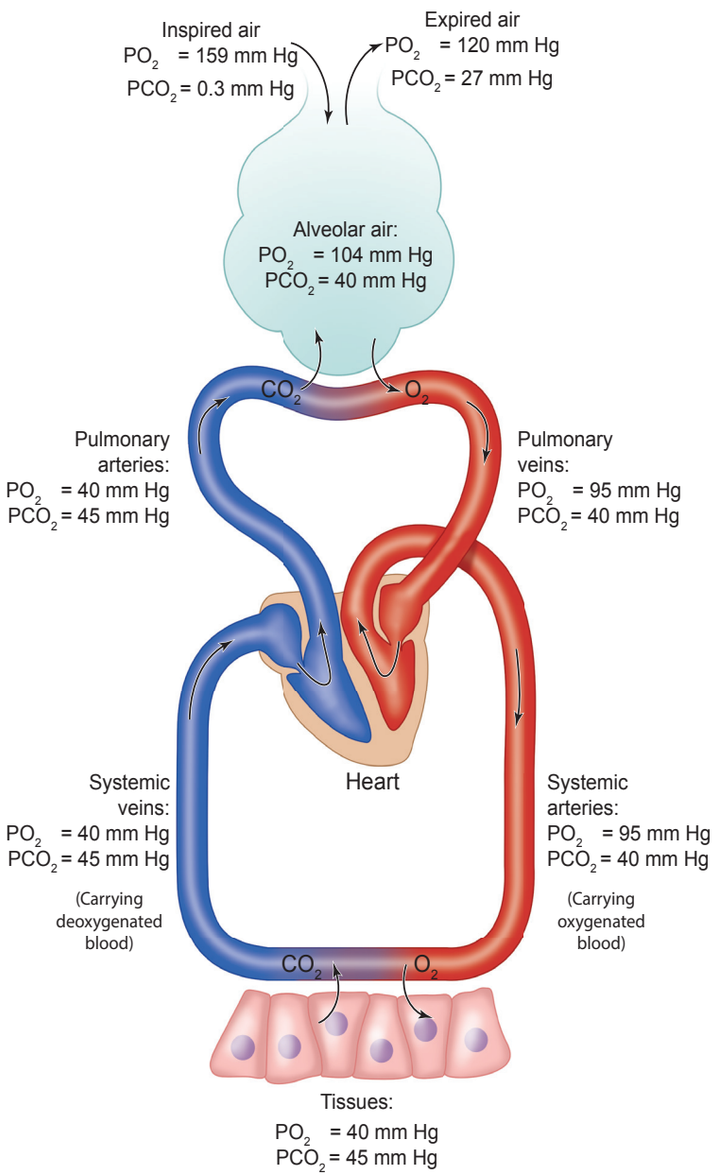


Figure 6.6 Exchange of gases at the alveolus and the tissue with blood and transport of oxygen and carbondioxide

XI Std Zoology Chapter-6 EM.indd 101 30/12/2022 10:41:56